Survival of the Fittest: A Mathematical Study of Multi-Strain SIR Dynamics under Differential Vaccine Efficacy

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Abstract

This project investigates the competitive dynamics of two distinct viral strains competing for a single host population. We extend the standard epidemiological framework to an SVIR (Susceptible-Vaccinated-Infected-Recovered) model to analyze how differential vaccine efficacy dictates which strain achieves endemicity and which is driven to extinction.

1 Introduction and Mathematical Model

We model the population using the following system of non-linear ordinary differential equations:

$$\frac{dS}{dt} = \Lambda - (\beta_1 I_1 + \beta_2 I_2)S - (\nu + \mu)S \tag{1}$$

$$\frac{dV}{dt} = \nu S - (\sigma_1 \beta_1 I_1 + \sigma_2 \beta_2 I_2)V - \mu V \tag{2}$$

$$\frac{dI_1}{dt} = (\beta_1 S + \sigma_1 \beta_1 V)I_1 - (\gamma_1 + \mu)I_1 \tag{3}$$

$$\frac{dI_2}{dt} = (\beta_2 S + \sigma_2 \beta_2 V)I_2 - (\gamma_2 + \mu)I_2 \tag{4}$$

$$\frac{dR}{dt} = \gamma_1 I_1 + \gamma_2 I_2 - \mu R \tag{5}$$

where $\sigma_n = 1 - \epsilon_n$, representing the vaccine leakage for strain n. We assume a constant population normalized to $N = 1$. Therefore, the recruitment rate is equal to the natural mortality rate ($\Lambda = \mu$), ensuring that $S + V + I_1 + I_2 + R = 1$ at all times t.

2 Local Stability Analysis

2.1 Disease-Free Equilibrium (DFE)

The Disease-Free Equilibrium (DFE) represents the state where no disease is present in the population. We find the DFE by setting the derivatives of the system to zero and assuming $I_1 = 0$ and $I_2 = 0$.

From Equation (1), setting $I_1 = I_2 = 0$:

$$\Lambda - (\nu + \mu)S = 0$$
$$S^* = \frac{\Lambda}{\nu + \mu}$$

From Equation (2):

$$\nu S^* - \mu V = 0$$
$$V^* = \frac{\nu}{\mu} S^* = \frac{\nu \Lambda}{\mu(\nu + \mu)}$$

From Equation (5), setting $I_1 = I_2 = 0$:

$$-\mu R = 0 \implies R^* = 0$$

Thus, the Disease-Free Equilibrium, denoted as E_0, is given by:

$$E_0 = \left(\frac{\Lambda}{\nu + \mu}, \frac{\nu \Lambda}{\mu(\nu + \mu)}, 0, 0, 0 \right) \tag{6}$$

2.2 Jacobian Matrix and Local Stability

To analyze the local stability of the DFE, we construct the Jacobian matrix J of the system. The matrix is given by:

$$J = \begin{bmatrix} -(\beta_1 I_1 + \beta_2 I_2) - (\nu + \mu) & 0 & -\beta_1 S & -\beta_2 S & 0 \\ \nu & -(\sigma_1 \beta_1 I_1 + \sigma_2 \beta_2 I_2) - \mu & -\sigma_1 \beta_1 V & -\sigma_2 \beta_2 V & 0 \\ \beta_1 I_1 & \sigma_1 \beta_1 I_1 & (\beta_1 S + \sigma_1 \beta_1 V) - (\gamma_1 + \mu) & 0 & 0 \\ \beta_2 I_2 & \sigma_2 \beta_2 I_2 & 0 & (\beta_2 S + \sigma_2 \beta_2 V) - (\gamma_2 + \mu) & 0 \\ 0 & 0 & \gamma_1 & \gamma_2 & -\mu \end{bmatrix} \tag{7}$$

Evaluating the Jacobian at the Disease-Free Equilibrium $E_0(S^*, V^*, 0, 0, 0)$:

$$J(E_0) = \begin{bmatrix} -(\nu + \mu) & 0 & -\beta_1 S^* & -\beta_2 S^* & 0 \\ \nu & -\mu & -\sigma_1 \beta_1 V^* & -\sigma_2 \beta_2 V^* & 0 \\ 0 & 0 & (\beta_1 S^* + \sigma_1 \beta_1 V^*) - (\gamma_1 + \mu) & 0 & 0 \\ 0 & 0 & 0 & (\beta_2 S^* + \sigma_2 \beta_2 V^*) - (\gamma_2 + \mu) & 0 \\ 0 & 0 & \gamma_1 & \gamma_2 & -\mu \end{bmatrix} \tag{8}$$

Because $J(E_0)$ is a block-triangular matrix, the eigenvalues are simply the entries on the main diagonal of the blocks: $\lambda_1 = -(\nu + \mu)$, $\lambda_2 = -\mu$, $\lambda_5 = -\mu$, and the two key eigenvalues:

$$\lambda_3 = (\beta_1 S^* + \sigma_1 \beta_1 V^*) - (\gamma_1 + \mu) \tag{9}$$
$$\lambda_4 = (\beta_2 S^* + \sigma_2 \beta_2 V^*) - (\gamma_2 + \mu) \tag{10}$$

For local asymptotic stability, we require all eigenvalues to be strictly negative ($\lambda_i < 0$). This directly leads to the derivation of the Basic Reproduction Number, R_0.

2.3 Basic Reproduction Number (R_0)

We can formally derive R_0 using the Next-Generation Matrix method. We isolate the infected compartments, I_1 and I_2. The rate of appearance of new infections ($\mathcal{F}$) and the rate of transfer out of the compartments ($\mathcal{V}$) are:

$$\mathcal{F} = \begin{bmatrix} (\beta_1 S + \sigma_1 \beta_1 V) I_1 \\ (\beta_2 S + \sigma_2 \beta_2 V) I_2 \end{bmatrix}, \quad \mathcal{V} = \begin{bmatrix} (\gamma_1 + \mu) I_1 \\ (\gamma_2 + \mu) I_2 \end{bmatrix} \tag{11}$$

The Jacobians F and V evaluated at the DFE $(S^*, V^*, 0, 0, 0)$ are:

$$F = \begin{bmatrix} \beta_1 S^* + \sigma_1 \beta_1 V^* & 0 \\ 0 & \beta_2 S^* + \sigma_2 \beta_2 V^* \end{bmatrix}, \quad V = \begin{bmatrix} \gamma_1 + \mu & 0 \\ 0 & \gamma_2 + \mu \end{bmatrix} \tag{12}$$

The Next-Generation Matrix is $K = FV^{-1}$:

$$K = \begin{bmatrix} \frac{\beta_1 S^* + \sigma_1 \beta_1 V^*}{\gamma_1 + \mu} & 0 \\ 0 & \frac{\beta_2 S^* + \sigma_2 \beta_2 V^*}{\gamma_2 + \mu} \end{bmatrix} \tag{13}$$

The basic reproduction number R_0 is the spectral radius of K, $\rho(K)$. Since K is a diagonal matrix, its eigenvalues are its diagonal entries. Thus, we can define the strain-specific reproduction numbers:

$$R_{0,1} = \frac{\beta_1 S^* + \sigma_1 \beta_1 V^*}{\gamma_1 + \mu} \tag{14}$$

$$R_{0,2} = \frac{\beta_2 S^* + \sigma_2 \beta_2 V^*}{\gamma_2 + \mu} \tag{15}$$

The overall basic reproduction number is $R_0 = \max(R_{0,1}, R_{0,2})$. If $R_0 < 1$, the DFE is locally asymptotically stable, meaning the disease will die out if introduced in small quantities.

3 Global Stability Analysis

To prove the global asymptotic stability (GAS) of the Disease-Free Equilibrium $E_0(S^*, V^*, 0, 0, 0)$ when $R_0 \leq 1$, we construct a Volterra-type Lyapunov function. Let $L : \mathbb{R}^4_+ \to \mathbb{R}$ be defined as:

$$L(S, V, I_1, I_2) = \left(S - S^* - S^* \ln \frac{S}{S^*}\right) + \left(V - V^* - V^* \ln \frac{V}{V^*}\right) + I_1 + I_2 \tag{16}$$

Note that L is positive definite for all $S, V > 0$ and $L = 0$ strictly at the equilibrium $(S^*, V^*, 0, 0)$. The orbital derivative of L along the solutions of the system is:

$$\frac{dL}{dt} = \left(1 - \frac{S^*}{S}\right) \frac{dS}{dt} + \left(1 - \frac{V^*}{V}\right) \frac{dV}{dt} + \frac{dI_1}{dt} + \frac{dI_2}{dt} \tag{17}$$

At the DFE, the following equilibrium identities hold: $\Lambda = (\nu + \mu) S^*$ and $\nu S^* = \mu V^*$. We substitute the ODEs into our derivative and use these identities:

$$\frac{dL}{dt} = \left(1 - \frac{S^*}{S}\right)[(\nu+\mu)(S^* - S) - (\beta_1 I_1 + \beta_2 I_2)S]$$
$$+ \left(1 - \frac{V^*}{V}\right)\left[\nu S - \frac{\nu S^*}{V^*}V - (\sigma_1\beta_1 I_1 + \sigma_2\beta_2 I_2)V\right]$$
$$+ [(\beta_1 S + \sigma_1\beta_1 V)I_1 - (\gamma_1+\mu)I_1] + [(\beta_2 S + \sigma_2\beta_2 V)I_2 - (\gamma_2+\mu)I_2]$$

Expanding the terms and grouping the I_1 and I_2 components:

$$\frac{dL}{dt} = -(\nu+\mu)\frac{(S-S^*)^2}{S} - (S - S^*)(\beta_1 I_1 + \beta_2 I_2)$$
$$+ \nu\left(S - S^*\frac{V}{V^*} - S\frac{V^*}{V} + S^*\right) - (V - V^*)(\sigma_1\beta_1 I_1 + \sigma_2\beta_2 I_2)$$
$$+ \beta_1 S I_1 + \sigma_1\beta_1 V I_1 - (\gamma_1+\mu)I_1 + \beta_2 S I_2 + \sigma_2\beta_2 V I_2 - (\gamma_2+\mu)I_2$$

Notice that the non-linear terms $(\beta_1 S I_1)$ and $(\sigma_1\beta_1 V I_1)$ cancel out perfectly with the respective expanded terms from the susceptible and vaccinated equations. Grouping the remaining I_1 and I_2 terms gives:

$$I_1[\beta_1 S^* + \sigma_1\beta_1 V^* - (\gamma_1+\mu)] = I_1(\gamma_1+\mu)(R_{0,1} - 1)$$
$$I_2[\beta_2 S^* + \sigma_2\beta_2 V^* - (\gamma_2+\mu)] = I_2(\gamma_2+\mu)(R_{0,2} - 1)$$

Now, we handle the non-infected terms. We split $-(\nu+\mu)\frac{(S-S^*)^2}{S}$ into $-\mu\frac{(S-S^*)^2}{S} - \nu\frac{(S-S^*)^2}{S}$ and group the ν components:

$$-\nu\left(S - 2S^* + \frac{(S^*)^2}{S}\right) + \nu\left(S - S^*\frac{V}{V^*} - S\frac{V^*}{V} + S^*\right) = \nu S^*\left(3 - \frac{S^*}{S} - \frac{V}{V^*} - \frac{SV^*}{S^*V}\right)$$

Thus, the fully simplified derivative is:

$$\frac{dL}{dt} = -\mu\frac{(S-S^*)^2}{S} + \nu S^*\left(3 - \frac{S^*}{S} - \frac{V}{V^*} - \frac{SV^*}{S^*V}\right) + (\gamma_1+\mu)(R_{0,1}-1)I_1 + (\gamma_2+\mu)(R_{0,2}-1)I_2 \tag{18}$$

Applying the AM-GM inequality to the three positive terms $\frac{S^*}{S}$, $\frac{V}{V^*}$, and $\frac{SV^*}{S^*V}$ yields an arithmetic mean that is greater than or equal to their geometric mean (which is 1). Thus:

$$3 - \frac{S^*}{S} - \frac{V}{V^*} - \frac{SV^*}{S^*V} \leq 0 \tag{19}$$

Therefore, if $R_{0,1} \leq 1$ and $R_{0,2} \leq 1$ (which implies $R_0 \leq 1$), every term in the expression for $\frac{dL}{dt}$ is less than or equal to zero.

Thus, $\frac{dL}{dt} \leq 0$. Furthermore, $\frac{dL}{dt} = 0$ strictly when $S = S^*$, $V = V^*$, $I_1 = 0$, and $I_2 = 0$. By LaSalle's Invariance Principle, the largest invariant set within $\{(S, V, I_1, I_2) \mid \frac{dL}{dt} = 0\}$ is the singleton $\{E_0\}$. Hence, the Disease-Free Equilibrium is Globally Asymptotically Stable when $R_0 \leq 1$.

Note: This Lyapunov function establishes global stability strictly for the disease-free state ($R_0 \leq 1$). Analyzing the global stability of the Endemic Equilibrium ($R_0 > 1$) requires a highly complex formulation outside the scope of this paper; hence, the behavior of the system for $R_0 > 1$ is explored numerically in Section 4.

4 Computational Simulation and Sensitivity Analysis

To bridge the theoretical mathematical analysis with applied epidemiological data, we performed numerical integration of the SVIR system using `scipy.integrate.odeint` in Python. We use an RK45-based solver to track the competition between two strains (e.g., Delta vs. Omicron) over a 1095-day (3-year) period. This specific timeframe was chosen to reflect realistic viral mutation timelines, capturing the acute displacement phase of the variants and their subsequent stabilization into an endemic state.

4.1 Parameters and Initial Conditions

The simulation relies on the following fixed parameters and initial conditions:

Parameter/Variable	Description	Value
μ	Natural mortality rate (assuming 70-year lifespan)	$1/(70 \times 365)$
Λ	Recruitment rate ($\Lambda = \mu$)	μ
γ_1, γ_2	Recovery rates for both strains	1/14
$S(0)$	Initial susceptible population fraction	0.948
$V(0)$	Initial vaccinated population fraction	0.05
$I_1(0), I_2(0)$	Initial infected population fractions	0.001
$R(0)$	Initial recovered population fraction	0.0

Table 1: Fixed parameters and initial conditions for the numerical simulation.

4.2 Scenario A: Baseline Competition

In our baseline scenario, we model a population where vaccination occurs at a standard rate ($\nu = 0.01$). Strain 1 represents a less evasive variant with a vaccine leakage of $\sigma_1 = 0.10$ (90% efficacy). Strain 2 represents a highly evasive variant with a leakage of $\sigma_2 = 0.60$ (40% efficacy) and a higher transmission rate ($\beta_2 = 1.16$ vs $\beta_1 = 0.60$).

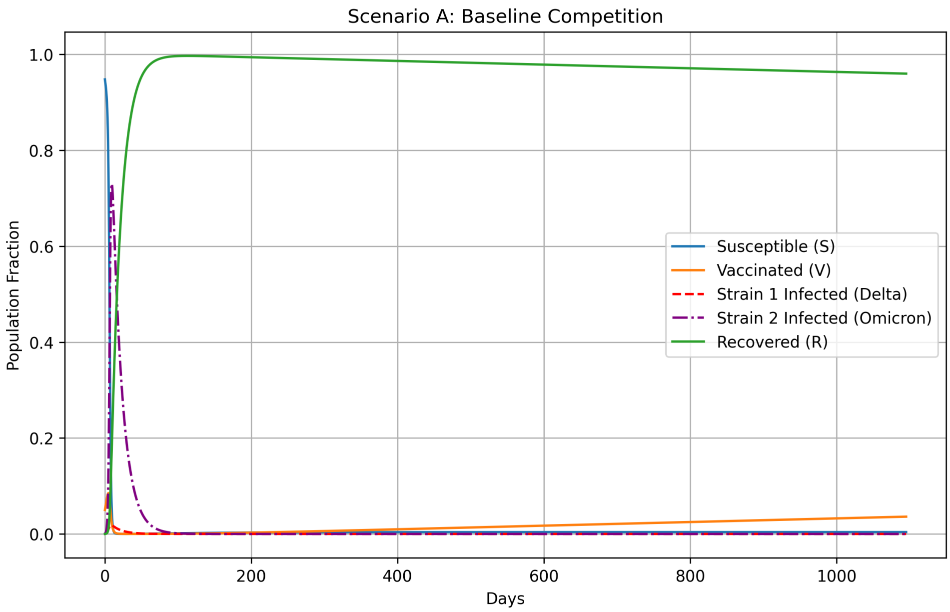

Figure 1: Baseline Competition showing Strain 2 driving Strain 1 to extinction.

As shown in Figure 1, Strain 1 peaks early due to the highly susceptible initial population. However, because of the higher immune escape (σ_2) and transmission rate (β_2), Strain 2 eventually dominates the host population, driving Strain 1 to extinction. This demonstrates the Competitive Exclusion Principle mathematically occurring within the host population.

Note: Because the demographic turnover (μ) is exceedingly slow (70-year lifespan), the virus rapidly exhausts the susceptible pool. The subsequent endemic equilibrium exists at a microscopic prevalence level, appearing as visual extinction on a linear scale, though mathematically $I_2 > 0$ for $t \to \infty$.

4.3 Scenario B: Vaccination Surge

We then increased the vaccination rate to $\nu = 0.05$ to observe the impact of a massive vaccination campaign.

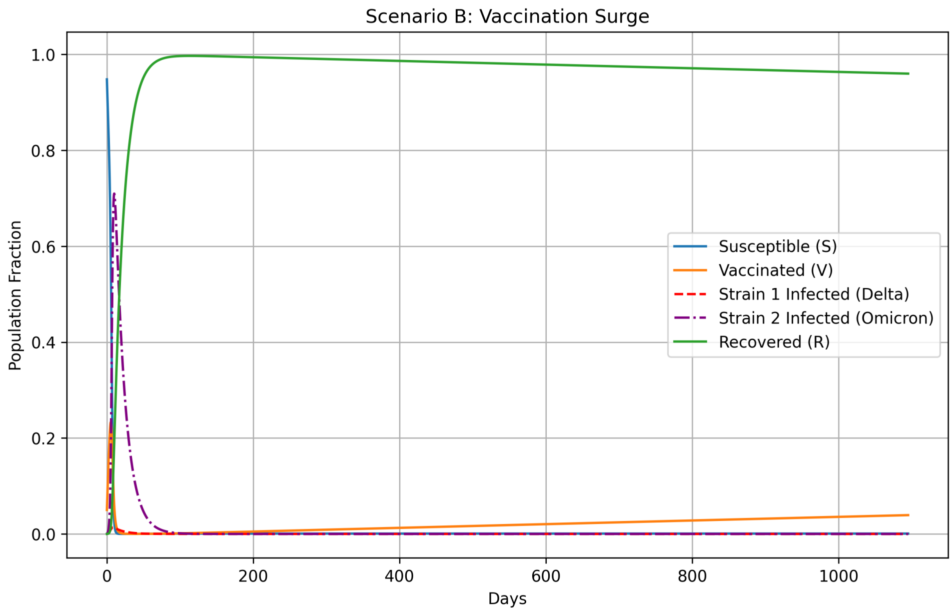

Figure 2: Impact of a Vaccination Surge.

In Figure 2, the rapid increase in the vaccinated population completely flattens the curve for Strain 1, effectively suppressing the outbreak. However, Strain 2 still experiences a significant surge. This proves that while increasing the vaccination rate protects against standard strains, it cannot fully stop a highly evasive variant.

4.4 Scenario C: Sensitivity of Strain 2 to Transmission Rate

To systematically analyze the exact threshold where the evasive variant overtakes the system, we ran simulations across 20 different values of β_2 (ranging from 0.5 to 1.5) and plotted the peak prevalence of the Strain 2 outbreak.

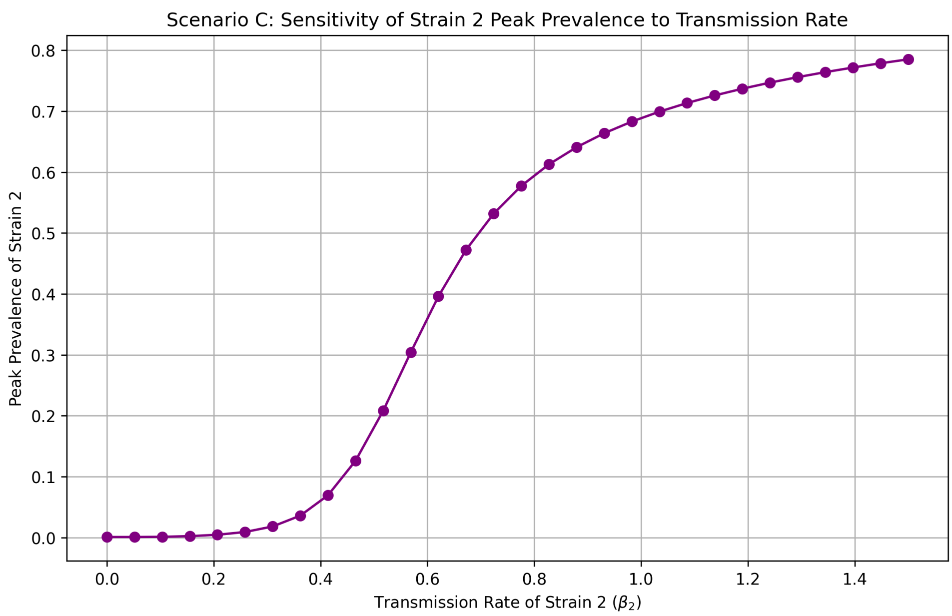

Figure 3: Sensitivity Analysis: Peak Prevalence vs Transmission Rate (β_2).

Figure 3 illustrates a profound phase transition driven by multi-strain competition. While our analytical derivation proves Strain 2 becomes viable at $\beta_2^* \approx 0.073$, the graph shows peak prevalence remaining near zero until approximately $\beta_2 \approx 0.45$. This discrepancy highlights the effect of resource depletion: in the range $0.073 < \beta_2 < 0.585$, Strain 2 is viable, but the faster Strain 1 ($\beta_1 = 0.60$) rapidly consumes the susceptible pool, artificially suppressing Strain 2's peak. The macroscopic exponential surge in Figure 3 only occurs as β_2 approaches our derived competitive dominance threshold ($\beta_2 \approx 0.585$), where Strain 2 becomes fast enough to outcompete Strain 1 for the available hosts.

While Figure 3 visually demonstrates the phase transition, we can analytically verify the exact tipping points using the Effective Reproduction Number (R_{eff}) at $t = 0$. For an outbreak to occur, we require $R_{eff,2}(0) > 1$, which algebraically simplifies to the critical threshold:

$$\beta_2^* = \frac{\gamma_2 + \mu}{S(0) + \sigma_2 V(0)}$$

Substituting our initial conditions yields $\beta_2^* \approx 0.073$. Below this transmission rate, Strain 2 immediately decays. Furthermore, for Strain 2 to outcompete Strain 1 during the initial acute phase, we require $R_{eff,2}(0) > R_{eff,1}(0)$. Setting the effective reproduction numbers equal and solving for β_2 yields a competitive dominance threshold of $\beta_2 \approx 0.585$. This analytical derivation perfectly mirrors our computational findings.

5 Conclusion

This project successfully modeled the multi-strain SIR dynamics under differential vaccine efficacy. Through rigorous mathematical analysis, we established the conditions for the Disease-Free Equilibrium and computed the Basic Reproduction Number (R_0) using the Next-Generation Matrix method. We further proved the Global Asymptotic Stability of the DFE using a Volterra-type Lyapunov function and LaSalle's Invariance Principle. Finally, our numerical simulations verified the Competitive Exclusion Principle. We analytically derived and computationally confirmed the exact transmission threshold ($\beta_2 \approx 0.073$) required for initial outbreak viability, as well as the threshold ($\beta_2 > 0.585$) required for acute competitive dominance.